Retail Sales Analytics Data Pipeline

Building a Modern Cloud-Based ELT Platform
using Airflow, Snowflake and dbt

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1. Executive Summary

Retail organizations generate large volumes of transactional sales data from multiple operational systems. However, fragmented data sources and manual processing create delays in reporting and decision-making.

This project presents a modern cloud-based ELT pipeline that automates the complete flow of retail sales data from ingestion to analytics. The solution uses Apache Airflow for orchestration, Snowflake for cloud data warehousing, dbt for transformation, Docker for containerization, and Databricks for dashboard visualization.

The pipeline improves data reliability, reduces manual effort, and provides analytics-ready datasets for business intelligence and KPI reporting.

2. Business Problem

Current Challenges

- Data scattered across multiple systems
- Manual ETL processes causing delays
- Inconsistent KPIs across departments
- Limited scalability for growing data volumes
- Difficulty maintaining data quality

Business Impact

Without a centralized data platform, stakeholders receive outdated and inconsistent insights. This affects sales analysis, inventory planning, and strategic decision-making.

3. Project Objectives

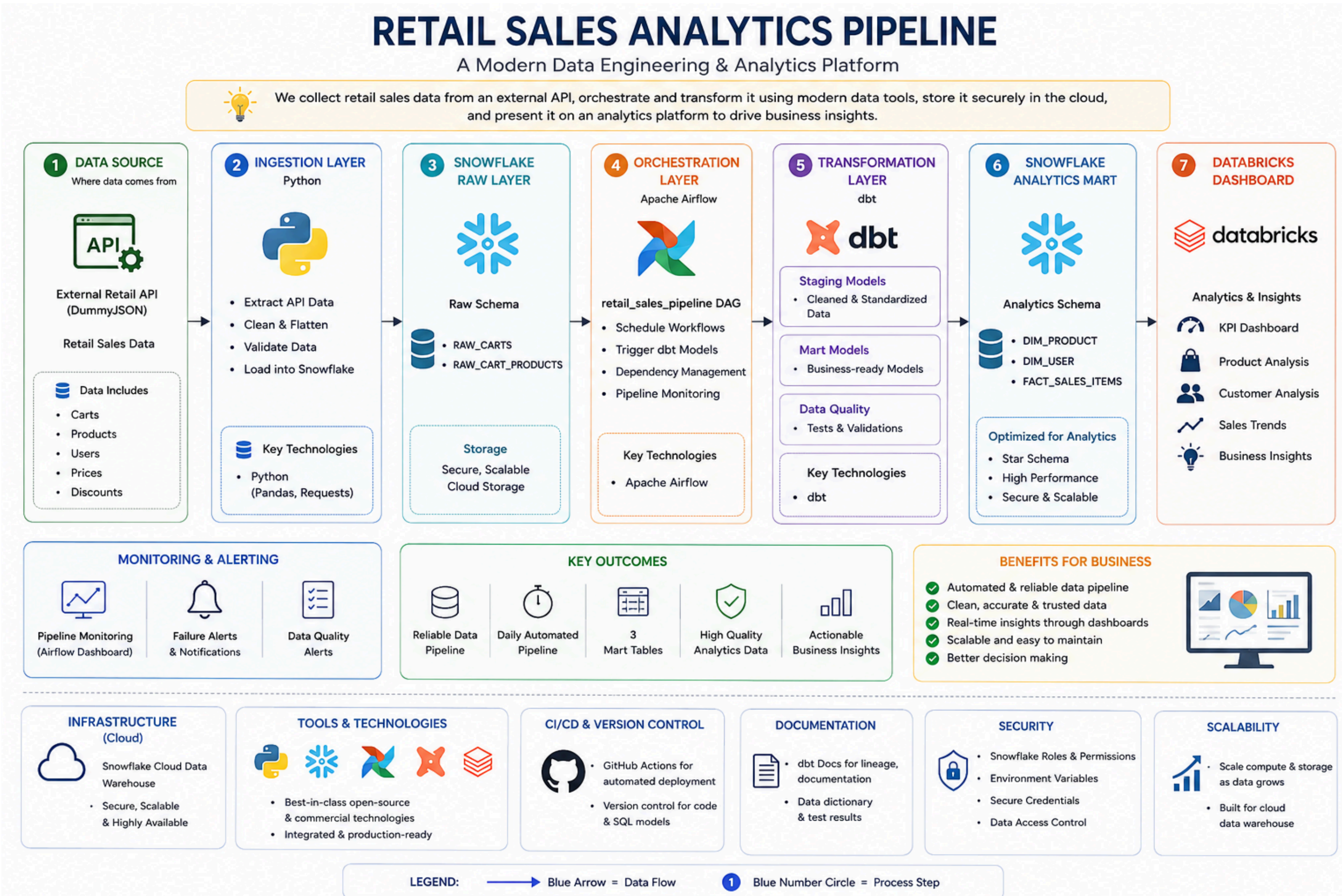
Core Objectives

- Automate retail sales data ingestion
- Build a scalable ELT pipeline
- Centralize data storage in Snowflake
- Improve data quality through validation
- Enable analytics-ready data models

Technical Objectives

- Apache Airflow - scheduling and orchestration
- dbt - modular transformations
- Docker containerize the solution
- Databricks - provide dashboard

4. Solution Architecture



Key Benefits

- Automated daily pipeline execution
- Scalable cloud data warehouse
- Reliable data validation
- Reduced operational overhead

5. Technology Stack

Layer	Technology	Purpose
Orchestration	Apache Airflow	Automates and schedules workflows
Programming Language	Python	Data ingestion and processing scripts
Data Warehouse	Snowflake	Scalable cloud-based analytics storage
Transformation	dbt (Data Build Tool)	SQL-based transformations and testing
Containerization	Docker	Ensures consistent runtime environments
Visualization	Databricks	Business intelligence dashboards and analytics
Version Control	GitHub	Code collaboration and version tracking

6. Pipeline Workflow & Data Quality Strategy

Pipeline Workflow

- Extract sales data from external API
- Load raw data into Snowflake
- Validate schema, data types, and duplicates
- Orchestrate workflow using Airflow
- Transform data using dbt
- Create fact and dimension tables
- Generate dashboards in Databricks

7. Data Quality Controls

Technology	Purpose
Schema Validation	Ensure correct structure
Null Checks	Prevent incomplete records
Duplicate Detection	Remove redundant transactions
Logging	Track pipeline execution
Retries	Recover from temporary failures
Failure Alerts	Notify on critical issues

8. Conclusion

The Retail Sales Analytics Data Pipeline demonstrates how modern cloud data engineering tools can be integrated to build an automated, scalable, and reliable ELT platform.

By combining Airflow, Snowflake, dbt, Docker, and Databricks, the solution transforms raw transactional data into analytics-ready insights with minimal manual intervention. The architecture improves data quality, reporting speed, operational efficiency, and business decision-making capabilities.